Name
Date
Course

# Formulas for Calculating Derivatives

## Alternate Notation for the Derivative

$$f'(x) = y' = \frac{dy}{dx} = \frac{d}{dx}[f(x)]$$

### Examples

$$\frac{d}{dx}(x^2) = 2x \qquad \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}} \qquad \frac{d}{dx}\left(\frac{1}{3x}\right) = -\frac{1}{3x^2}$$

## Differentiation Formulas

Constant Rule i) $\frac{d}{dx}(c) = 0$ "c" is a constant.

Power Rule ii) $\frac{d}{dx}(x^n) = nx^{n-1}$ "n" is a constant.

Constant Multiple Rule iii) $\frac{d}{dx}[cf(x)] = cf'(x)$ "c" is a constant.

Sum Rule iv) $\frac{d}{dx}[f(x)+g(x)] = f'(x) + g'(x)$

Difference Rule v) $\frac{d}{dx}[f(x)-g(x)] = f'(x) - g'(x)$

Product Rule vi) $\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x)$

Quotient Rule vii) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$

## A Proof of the Constant Rule

If $f(x) = c$, then "c" is a constant.

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{c-c}{h} = \lim_{h\to 0} \frac{0}{h} \overset{6}{=} \lim_{h\to 0} (0) = \boxed{0}$$

## A Proof of the Constant Multiple Rule

If $g(x) = cf(x)$, then

$$g'(x) = \lim_{h\to 0} \frac{g(x+h) - g(x)}{h} = \lim_{h\to 0} \frac{cf(x+h) - cf(x)}{h}$$

$$= \lim_{h\to 0} \left\{ \frac{c[f(x+h) - f(x)]}{h} \right\}$$

$$\overset{5}{=} c \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$$

$$= cf'(x)$$

# A Proof of the Power Rule

If $f(x)=x^n$, then $f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{(x+h)^n-x^n}{h}$

## The Binomial Theorem

$$(a+b)^n=\binom{n}{0}a^n b^0+\binom{n}{1}a^{n-1}b^1+\binom{n}{2}a^{n-2}b^2+\dots+\binom{n}{n}a^0 b^n$$

"a" and "b" must be real numbers and n must be a positive integer.

$\binom{n}{r}=\frac{n!}{r!(n-r)!}$

$$=\lim_{h\to 0}\frac{\binom{n}{0}x^n h^0+\binom{n}{1}x^{n-1}h^1+\binom{n}{2}x^{n-2}h^2+\dots+\binom{n}{n}x^0 h^n-x^n}{h}$$

$$=\lim_{h\to 0}\frac{(1)x^n(1)+nx^{n-1}h+\frac{n(n-1)}{2}x^{n-2}h^2+\dots+(1)(1)h^n-x^n}{h}$$

$$=\lim_{h\to 0}\frac{x^n+nx^{n-1}h+\frac{n(n-1)}{2}x^{n-2}h^2+\dots+h^n-x^n}{h}$$

$$=\lim_{h\to 0}\frac{nx^{n-1}h+\frac{n(n-1)}{2}x^{n-2}h^2+\dots+h^n}{h}$$

$$=\lim_{h\to 0}\left(nx^{n-1}+\frac{n(n-1)}{2}x^{n-2}h+\dots+h^{n-1}\right)$$

$$=nx^{n-1}+\frac{n(n-2)}{2}x^{n-2}(0)+\dots+0^{n-1}$$

$$=nx^{n-1}$$

- Find $f'(x)$.

① $f(x)=5$

$f'(x)\overset{\text{o}}{=}\boxed{0}$

② $f(x)=x^5$

$f'(x)\overset{\text{oo}}{=}5x^{5-1}=\boxed{5x^4}$

③ $f(x)=-3x^{12}$

$f'(x)\overset{\text{ooo}}{=}-3\frac{d}{dx}(x^{12})\overset{\text{oo}}{=}-3(12x^{12-1})=\boxed{-36x^{11}}$

Find $\frac{dy}{dx}$.

④ $y=\frac{2}{7}$

$\frac{dy}{dx}\overset{\text{o}}{=}\boxed{0}$

⑤ $y=x^6$

$\frac{dy}{dx}\overset{\text{oo}}{=}6x^{6-1}=\boxed{6x^5}$

⑥ $y = 10x^8$

$$\frac{dy}{dx} \overset{iii}{=} 10\frac{d}{dx}(x^8) \overset{ii}{=} 10(8x^{8-1}) = \boxed{80x^7}$$

Find $y'$.

⑦ $y = c^2$

$$y' \overset{i}{=} \boxed{0}$$

⑧ $y = x^9$

$$y' \overset{ii}{=} 9x^{9-1} = \boxed{9x^8}$$

⑨ $y = -7x^4$

$$y' \overset{iii}{=} -7\frac{d}{dx}(x^4) \overset{ii}{=} -7(4x^{4-1}) = \boxed{-28x^3}$$

Find $f'(x)$

⑩ $f(x) = 8$

$$f'(x) \overset{i}{=} \boxed{0}$$

⑪ $f(x) = x^{100}$

$$f'(x) \overset{ii}{=} 100x^{100-1} = \boxed{100x^{99}}$$

⑫ $f(x) = 5x^3$

$$f'(x) \overset{iii}{=} 5\frac{d}{dx}(x^3) \overset{ii}{=} 5(3x^2) = \boxed{15x^2}$$

Find $\frac{dy}{dx}$.

⑬ $y = -0.4$

$$\frac{dy}{dx} \overset{i}{=} \boxed{0}$$

⑭ $y = x^{11}$

$$\frac{dy}{dx} \overset{ii}{=} 11x^{11-1} = \boxed{11x^{10}}$$

⑮ $y = -8x^7$

$$\frac{dy}{dx} \overset{iii}{=} -8\frac{d}{dx}(x^7) \overset{ii}{=} -8(7x^{7-1}) = \boxed{-56x^6}$$

Find $y'$.

(16) $y = -6\sqrt{7} - 2$

$$y' \overset{i}{=} \boxed{0}$$

(17) $y = x^{29}$

$$y' \overset{ii}{=} \boxed{29x^{28}}$$

(18) $y = 2x^4$

$$y' \overset{iii}{=} 2\frac{d}{dx}(x^4) \overset{ii}{=} 2(4x^{4-1}) = \boxed{8x^3}$$

Find $f'(x)$.

(19) $f(x) = \dfrac{1}{x^5}$

$$f(x) = x^{-5}$$

$$f'(x) \overset{ii}{=} -5x^{-5-1} = -5x^{-6} = -5\left(\frac{1}{x^6}\right) = \boxed{-\frac{5}{x^6}}$$

Find $\dfrac{dy}{dx}$.

(20) $y = \dfrac{1}{x^{10}}$

$$y = x^{-10}$$

$$y' \overset{ii}{=} -10x^{-10-1} = -10x^{-11} = -10\left(\frac{1}{x^{11}}\right) = \boxed{\frac{-10}{x^{11}}}$$

Find $y'$.

(21) $y = \dfrac{-6}{x^8}$

$$y = -6\left(\frac{1}{x^8}\right) = -6x^{-8}$$

$$y' \overset{iii}{=} -6\frac{d}{dx}(x^{-8}) \overset{ii}{=} -6(-8x^{-8-1}) = 48x^{-9} = 48\left(\frac{1}{x^9}\right)$$

$$= \boxed{\frac{48}{x^9}}$$

Find $f'(x)$.

(22) $f(x) = \frac{2}{x^6}$

$f(x) = 2\left(\frac{1}{x^6}\right) = 2x^{-6}$

$f'(x) \overset{iii}{=} 2\frac{d}{dx}(x^{-6}) \overset{ii}{=} 2(-6x^{-6-1}) = -12x^{-7} = -12\left(\frac{1}{x^7}\right)$

$= \boxed{-\frac{12}{x^7}}$

Find $\frac{dy}{dx}$.

(23) $y = \frac{1}{6x^2}$

$y = \frac{1}{6}\left(\frac{1}{x^2}\right) = \frac{1}{6}x^{-2}$

$\frac{dy}{dx} \overset{iii}{=} \left(\frac{1}{6}\right)\frac{d}{dx}(x^{-2}) \overset{ii}{=} \frac{1}{6}(-2x^{-2-1}) = -\frac{1}{3}x^{-3} = -\frac{1}{3}\left(\frac{1}{x^3}\right)$

$= \boxed{-\frac{1}{3x^3}}$

Find $y'$

(24) $y = \frac{1}{2x^3}$

$y = \frac{1}{2}\left(\frac{1}{x^3}\right) = \frac{1}{2}x^{-3}$

$y' \overset{iii}{=} \left(\frac{1}{2}\right)\frac{d}{dx}(x^{-3}) \overset{ii}{=} \frac{1}{2}(-3x^{-3-1}) = -\frac{3}{2}x^{-4} = -\frac{3}{2}\left(\frac{1}{x^4}\right)$

$= \boxed{-\frac{3}{2x^4}}$

Find $f'(x)$

(25) $f(x) = \sqrt{x}$

$f(x) = x^{\frac{1}{2}}$

$f'(x) \overset{ii}{=} \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{-\frac{1}{2}} = \frac{1}{2}\left(\frac{1}{x^{1/2}}\right) = \frac{1}{2}\left(\frac{1}{\sqrt{x}}\right)$

$= \boxed{\frac{1}{2\sqrt{x}}}$

Find $\frac{dy}{dx}$.

(26) $y = \sqrt[6]{x}$

$$y = x^{\frac{1}{6}}$$

$$\frac{dy}{dx} = \frac{1}{6}x^{\frac{1}{6}-1} = \frac{1}{6}x^{-\frac{5}{6}} = \frac{1}{6}\left(\frac{1}{x^{5/6}}\right) = \frac{1}{6}\left(\frac{1}{\sqrt[6]{x^5}}\right) = \boxed{\frac{1}{6\sqrt[6]{x^5}}}$$

Find $y'$.

(27) $y = \sqrt[3]{x}$

$$y = x^{\frac{1}{3}}$$

$$y' = \frac{1}{3}x^{\frac{1}{3}-1} = \frac{1}{3}x^{-\frac{2}{3}} = \frac{1}{3}\left(\frac{1}{x^{2/3}}\right) = \frac{1}{3}\left(\frac{1}{\sqrt[3]{x^2}}\right) = \boxed{\frac{1}{3\sqrt[3]{x^2}}}$$

Find $f'(x)$

(28) $f(x) = -10\sqrt[4]{x^7}$

$$f(x) = -10x^{\frac{7}{4}}$$

$$f'(x) = -10\frac{d}{dx}\left(x^{\frac{7}{4}}\right) = -10\left(\frac{7}{4}x^{\frac{7}{4}-1}\right) = -\frac{70}{4}x^{\frac{3}{4}} = \boxed{-\frac{35}{2}\sqrt[4]{x^3}}$$

(29) $f(x) = 3\sqrt{x^3}$

$$f(x) = 3x^{\frac{3}{2}}$$

$$f'(x) = 3\frac{d}{dx}\left(x^{\frac{3}{2}}\right) = 3\left(\frac{3}{2}x^{\frac{3}{2}-1}\right) = \frac{9}{2}x^{\frac{1}{2}} = \boxed{\frac{9}{2}\sqrt{x}}$$

Find $\frac{dy}{dx}$. $(ab)^n = a^n b^n$

(30) $y = (3x)^3$

$$y = 3^3(x^3) = 27x^3$$

$$y' = 27\frac{d}{dx}(x^3) = 27(3x^{3-1}) = \boxed{81x^2}$$

(31) $y = (2x)^5$

$$y = 2^5x^5 = 32x^5$$

$$y' = 32\frac{d}{dx}(x^5) = 32(5x^{5-1}) = \boxed{160x^4}$$

Find $y'$! $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

(32) $y = \left(\frac{x}{7}\right)^2$

$$y = \frac{x^2}{7^2} = \frac{x^2}{49} = \frac{1}{49}x^2$$

$$y' \overset{iii}{=} \frac{1}{49}\frac{d}{dx}(x^2) \overset{ii}{=} \frac{1}{49}(2x^{2-1}) = \frac{1}{49}(2x^1) = \boxed{\frac{2}{49}x}$$

(33) $y = \left(\frac{x}{4}\right)^3$

$$y = \frac{x^3}{4^3} = \frac{x^3}{64} = \frac{1}{64}x^3$$

$$y' \overset{iii}{=} \frac{1}{64}\frac{d}{dx}(x^3) \overset{ii}{=} \frac{1}{64}(3x^{3-1}) = \boxed{\frac{3}{64}x^2}$$

Find $f'(x)$. $\sqrt[n]{ab} = \sqrt[n]{a} \cdot \sqrt[n]{b}$

(34) $f(x) = \sqrt[3]{5x}$

$$f(x) = \sqrt[3]{5} \cdot \sqrt[3]{x} = \sqrt[3]{5}\, x^{\frac{1}{3}}$$

$$f'(x) \overset{iii}{=} \sqrt[3]{5}\frac{d}{dx}\left(x^{\frac{1}{3}}\right) \overset{ii}{=} \sqrt[3]{5}\left(\frac{1}{3}x^{\frac{1}{3}-1}\right) = \frac{\sqrt[3]{5}}{3}x^{-\frac{2}{3}}$$

$$= \frac{\sqrt[3]{5}}{3}\left(\frac{1}{x^{2/3}}\right)$$

$$= \frac{\sqrt[3]{5}}{3}\frac{1}{\sqrt[3]{x^2}}$$

$$= \frac{\sqrt[3]{5}}{3\sqrt[3]{x^2}}$$

(35) $f(x) = \sqrt{7x}$

$$f(x) = \sqrt{7} \cdot \sqrt{x} = \sqrt{7}\, x^{\frac{1}{2}}$$

$$f'(x) \overset{iii}{=} \sqrt{7}\frac{d}{dx}\left(x^{\frac{1}{2}}\right) \overset{ii}{=} \sqrt{7}\left(\frac{1}{2}x^{\frac{1}{2}-1}\right) = \frac{\sqrt{7}}{2}x^{-\frac{1}{2}} = \frac{\sqrt{7}}{2}\left(\frac{1}{x^{1/2}}\right) = \frac{\sqrt{7}}{2}\left(\frac{1}{\sqrt{x}}\right)$$

$$= \boxed{\frac{\sqrt{7}}{2\sqrt{x}}}$$

Find $\frac{dy}{dx}$.

(36) $y = \dfrac{1}{\sqrt[4]{x^3}}$

$$y = \frac{1}{x^{\frac{3}{4}}} = x^{-\frac{3}{4}}$$

$$\frac{dy}{dx} \overset{ii}{=} -\frac{3}{4}x^{-\frac{3}{4}-1} = -\frac{3}{4}x^{-\frac{7}{4}} = -\frac{3}{4}\left(\frac{1}{x^{7/4}}\right) = -\frac{3}{4}\,\frac{1}{\sqrt[4]{x^7}} = \boxed{-\frac{3}{4\sqrt[4]{x^7}}}$$

(37) $y = \dfrac{-9}{(\sqrt[5]{x})^6}$

$$y = (-9)\frac{1}{(\sqrt[5]{x})^6} = -9\left(\frac{1}{x^{6/5}}\right) = -9x^{-\frac{6}{5}}$$

$$\frac{dy}{dx} \overset{iii}{=} -9\frac{d}{dx}x^{-\frac{6}{5}} \overset{ii}{=} -9\left(-\frac{6}{5}x^{-\frac{6}{5}-1}\right) = \frac{54}{5}x^{-\frac{11}{5}} = \frac{54}{5}\left(\frac{1}{x^{11/5}}\right)$$

$$= \frac{54}{5}\,\frac{1}{\sqrt[5]{x^{11}}}$$

$$= \boxed{\frac{54}{5\sqrt[5]{x^{11}}}}$$

Find $y'$.

(38) $y = \dfrac{1}{(\sqrt{x})^7}$

$$y = \frac{1}{x^{\frac{7}{2}}} = x^{-\frac{7}{2}}$$

$$y' \overset{ii}{=} -\frac{7}{2}x^{-\frac{7}{2}-1} = -\frac{7}{2}x^{-\frac{9}{2}} = -\frac{7}{2}\left(\frac{1}{x^{9/2}}\right) = -\frac{7}{2}\left(\frac{1}{\sqrt{x^9}}\right)$$

$$= \boxed{-\frac{7}{2\sqrt{x^9}}}$$

(39) $y = \dfrac{16}{\sqrt[8]{x^5}}$

$$y = 16\left(\frac{1}{\sqrt[8]{x^5}}\right) = 16\left(\frac{1}{x^{5/8}}\right) = 16x^{-\frac{5}{8}}$$

$$y' \overset{iii}{=} 16\frac{d}{dx}\left(x^{-\frac{5}{8}}\right) \overset{ii}{=} 16\left(-\frac{5}{8}x^{-\frac{5}{8}-1}\right) = -10x^{-\frac{13}{8}} = -10\left(\frac{1}{x^{13/8}}\right)$$

$$= -10\left(\frac{1}{\sqrt[8]{x^{13}}}\right)$$

$$= \boxed{-\frac{10}{\sqrt[8]{x^{13}}}}$$

Find $f'(x)$.

(40) $f(x) = \sqrt{\frac{2}{3}}$

$f'(x) \overset{i}{=} \boxed{0}$

$a^n \cdot a^m = a^{n+m}$

(41) $f(x) = x\sqrt{x}$

$f(x) = x^1 \cdot x^{\frac{1}{2}} = x^{1+\frac{1}{2}} = x^{\frac{3}{2}}$

$f'(x) \overset{ii}{=} \frac{3}{2}x^{\frac{3}{2}-1} = \frac{3}{2}x^{\frac{1}{2}} = \boxed{\frac{3}{2}\sqrt{x}}$

(42) $f(x) = 2x^3\sqrt[4]{x^5}$

$f(x) = 2x^3 \cdot x^{\frac{5}{4}} = 2x^{3+\frac{5}{4}} = 2x^{\frac{17}{4}}$

$f'(x) \overset{iii}{=} 2\frac{d}{dx}(x^{\frac{17}{4}}) \overset{ii}{=} 2\left(\frac{17}{4}x^{\frac{17}{4}-1}\right) = \frac{17}{2}x^{\frac{13}{4}} = \boxed{\frac{17}{2}\sqrt[4]{x^{13}}}$

(43) $f(x) = \frac{x}{\sqrt[3]{x}}$

$\frac{a^n}{a^m} = a^{n-m}$

$f(x) = \frac{x^1}{x^{\frac{1}{3}}} = x^{1-\frac{1}{3}} = x^{\frac{2}{3}}$

$f'(x) \overset{ii}{=} \frac{2}{3}x^{\frac{2}{3}-1} = \frac{2}{3}x^{-\frac{1}{3}} = \frac{2}{3}\left(\frac{1}{x^{1/3}}\right) = \frac{2}{3}\left(\frac{1}{\sqrt[3]{x}}\right) = \boxed{\frac{2}{3\sqrt[3]{x}}}$

Find $\frac{dy}{dx}$.

(44) $y = \pi^8$

$\frac{dy}{dx} \overset{i}{=} \boxed{0}$

(45) $y = x^2\sqrt[5]{x}$

$y = x^2 \cdot x^{\frac{1}{5}} = x^{2+\frac{1}{5}} = x^{\frac{11}{5}}$

$\frac{dy}{dx} \overset{ii}{=} \frac{11}{5}x^{\frac{11}{5}-1} = \frac{11}{5}x^{\frac{6}{5}} = \boxed{\frac{11}{5}\sqrt[5]{x^6}}$

(46) $y = x^5\sqrt{x^7}$

$y = x^5 \cdot x^{\frac{7}{2}} = x^{5+\frac{7}{2}} = x^{\frac{17}{2}}$

$\frac{dy}{dx} \overset{ii}{=} \frac{17}{2}x^{\frac{17}{2}-1} = \frac{17}{2}x^{\frac{15}{2}} = \boxed{\frac{17}{2}\sqrt{x^{15}}}$

(47) $y = \frac{\sqrt{x}}{x}$

$y = \frac{x^{\frac{1}{2}}}{x^1} = x^{\frac{1}{2}-1} = x^{-\frac{1}{2}}$

$\frac{dy}{dx} \overset{ii}{=} -\frac{1}{2}x^{-\frac{1}{2}-1} = -\frac{1}{2}x^{-\frac{3}{2}} = -\frac{1}{2}\left(\frac{1}{x^{3/2}}\right) = -\frac{1}{2}\left(\frac{1}{\sqrt{x^3}}\right) = \boxed{-\frac{1}{2\sqrt{x^3}}}$

Find $y'$.

$$\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$$

(48) $y = \sqrt{\frac{x}{6}}$

$$y = \frac{\sqrt{x}}{\sqrt{6}} = \frac{1}{\sqrt{6}}(\sqrt{x}) = \frac{1}{\sqrt{6}}\left(x^{\frac{1}{2}}\right)$$

$$\sqrt[n]{ab} = \sqrt[n]{a}\cdot\sqrt[n]{b}$$

$$y' \overset{iii}{=} \frac{1}{\sqrt{6}}\frac{d}{dx}\left(x^{\frac{1}{2}}\right) \overset{ii}{=} \frac{1}{\sqrt{6}}\left(\frac{1}{2}x^{\frac{1}{2}-1}\right) = \frac{1}{2\sqrt{6}}x^{-\frac{1}{2}} = \frac{1}{2\sqrt{6}}\left(\frac{1}{x^{1/2}}\right)$$

$$= \frac{1}{2\sqrt{6}}\left(\frac{1}{\sqrt{x}}\right)$$

$$= \boxed{\frac{1}{2\sqrt{6x}}}$$

(49) $y = \sqrt[6]{\frac{x}{2}}$

$$y = \frac{\sqrt[6]{x}}{\sqrt[6]{2}} = \frac{1}{\sqrt[6]{2}}(\sqrt[6]{x}) = \frac{1}{\sqrt[6]{2}}\left(x^{\frac{1}{6}}\right)$$

$$y' \overset{iii}{=} \frac{1}{\sqrt[6]{2}}\frac{d}{dx}\left(x^{\frac{1}{6}}\right) \overset{ii}{=} \frac{1}{\sqrt[6]{2}}\left(\frac{1}{6}x^{\frac{1}{6}-1}\right) = \frac{1}{6\sqrt[6]{2}}x^{-\frac{5}{6}} = \frac{1}{6\sqrt[6]{2}}\left(\frac{1}{x^{5/6}}\right)$$

$$= \frac{1}{6\sqrt[6]{2}}\,\frac{1}{\sqrt[6]{x^5}}$$

$$= \boxed{\frac{1}{6\sqrt[6]{2x^5}}}$$

Find $f'(x)$.

(50) $f(x) = x$

$$f(x) = x^1$$

$$f'(x) \overset{ii}{=} 1x^{1-1} = 1x^0 = 1(1) = \boxed{1}$$

(51) $f(x) = 8x$

$$f(x) = 8x^1$$

$$f'(x) \overset{iii}{=} 8\frac{d}{dx}(x^1) \overset{ii}{=} 8(1x^{1-1}) = 8x^0 = 8(1) = \boxed{8}$$

Find $\frac{dy}{dx}$.

(52) $y = -7x$

$$y = -7x^1$$

$$\frac{dy}{dx} \overset{iii}{=} -7\frac{d}{dx}(x^1) \overset{ii}{=} -7(1x^{1-1}) = -7x^0 = -7(1) = \boxed{-7}$$

(53) $y = \frac{1}{2}x$

$$y = \frac{1}{2}x^1$$

$$\frac{dy}{dx} \overset{iii}{=} \frac{1}{2}\frac{d}{dx}(x^1) \overset{ii}{=} \frac{1}{2}(1x^{1-1}) = \frac{1}{2}x^0 = \frac{1}{2}(1) = \boxed{\frac{1}{2}}$$

# Differentiation Formulas #4 & #5

## A Proof of the Sum Rule

If $r(x) = f(x) + g(x)$, then $r'(x) = \lim_{h\to 0} \frac{r(x+h) - r(x)}{h}$

Example: $2x^3 + 4x$
$f(x) + g(x)$

$$= \lim_{h\to 0} \frac{f(x+h) + g(x+h) - [f(x) + g(x)]}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h) + g(x+h) - f(x) - g(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x+h) - f(x) + g(x+h) - g(x)}{h}$$

$$= \lim_{h\to 0} \left[\frac{f(x+h) - f(x)}{h} + \frac{g(x+h) - g(x)}{h}\right]$$

$$\overset{!}{=} \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} + \lim_{h\to 0} \frac{g(x+h) - g(x)}{h}$$

$$= f'(x) + g'(x)$$

Find $f'(x)$.

(54) $f(x) = x^3 + 2x^5$

$f'(x) = \boxed{3x^2 + 10x^4}$

(55) $f(x) = -8x^4 + x^6 - 7$

$f'(x) = -32x^3 + 6x^5 + 0$

$f'(x) = \boxed{-32x^3 + 6x^5}$

(56) $f(x) = x^9 + x^8 + 5x^7 - \frac{1}{6}x^6 + 22$

$f'(x) = \boxed{9x^8 + 8x^7 + 35x^6 - x^5}$

Find $\frac{dy}{dx}$.

(57) $y = 3x^5 - 2x^3 + 1$

$\frac{dy}{dx} = \boxed{15x^4 - 6x^2}$

(58) $y = -x^4 + 3x^3 - \pi$

$\frac{dy}{dx} = \boxed{-4x^3 + 9x^2}$

(59) $y = 9x^8 - \frac{1}{2}x^4 + 2\sqrt{5}$

$\frac{dy}{dx} = \boxed{72x^7 - 2x^3}$

Find $y'$.

(60) $y = 2\sqrt[7]{x} + \frac{1}{x^6} + c^3$

$y = 2x^{\frac{1}{7}} + x^{-6} + c^3$

$y' = \frac{2}{7}x^{-\frac{6}{7}} - 6x^{-7} = \boxed{\frac{2}{7\sqrt[7]{x^6}} - \frac{6}{x^7}}$

(61) $y = \frac{3}{x^2} - \frac{1}{5x^3} - \sqrt[4]{x^9} - 10$

$y = 3x^{-2} - \frac{1}{5}x^{-3} - x^{\frac{9}{4}} - 10$

$y' = -6x^{-3} + \frac{3}{5}x^{-4} - \frac{9}{4}x^{\frac{5}{4}} = \boxed{-\frac{6}{x^3} + \frac{3}{5x^4} - \frac{9}{4}\sqrt[4]{x^5}}$

Find $f'(x)$.

(62) $f(x) = \frac{1}{\sqrt[3]{x^8}} + (5x)^2$

$f(x) = x^{-\frac{8}{3}} + 25x^2$

$f'(x) = -\frac{8}{3}x^{-\frac{11}{3}} + 50x = \boxed{-\frac{8}{3\sqrt[3]{x^{11}}} + 50x}$

(63) $f(x) = \sqrt[4]{\frac{x}{3}} + \frac{9}{x^4} + \sqrt{\frac{7}{3}}$

$f(x) = \frac{1}{\sqrt[4]{3}}x^{\frac{1}{4}} + 9x^{-4} + \sqrt{\frac{7}{3}}$

$f'(x) = \frac{1}{4\sqrt[4]{3}}x^{-\frac{3}{4}} - 36x^{-5} = \boxed{\frac{1}{4\sqrt[4]{3}\sqrt[4]{x^3}} - \frac{36}{x^5}}$

Find $\frac{dy}{dx}$.

(64) $y = \frac{x^5}{2} - \frac{x^4}{4}$

$y = \frac{1}{2}x^5 - \frac{1}{4}x^4$

$\frac{dy}{dx} = \boxed{\frac{5}{2}x^4 - x^3}$

(65) $y = \frac{x^6}{7} - \frac{5x^2}{3}$

$y = \frac{1}{7}x^6 - \frac{5}{3}x^2$

$\frac{dy}{dx} = \boxed{\frac{6}{7}x^5 - \frac{10}{3}x}$

(66) $y = \frac{x^{12}}{3} - \frac{x^8}{8} + \frac{7x^5}{3}$

$y = \frac{1}{3}x^{12} - \frac{1}{8}x^8 + \frac{7}{3}x^5$

$\frac{dy}{dx} = \boxed{4x^{11} - x^7 + \frac{35}{3}x^4}$

(67) $y = \frac{x^9}{3} + \frac{x^5}{5} - \frac{x^7}{2}$

$y = \frac{1}{3}x^9 + \frac{1}{5}x^5 - \frac{1}{2}x^7$

$\frac{dy}{dx} = \boxed{3x^8 + x^4 - \frac{7}{2}x^6}$